An Exact Extremal Formula for Triangle-Free Graphs with Bounded Degree and Matching Number

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Abstract

We determine the maximum number of edges in a triangle-free graph with prescribed upper bounds on maximum degree and matching number. The result is an explicit formula for all degrees $d \geq 2$ and matching bounds $m \geq 0$, together with constructions attaining it. We combine the component reduction of Ahanjideh, Ekim, and Yıldız and the construction family of Banak, Ekim, and Taşkın with a new fixed-order bound. Its proof extracts a saturated five-cycle, forces a homomorphism to C_5 under a total-deficiency hypothesis, and uses an odd-excess imbalance to bound the missing edges. The same argument determines the remaining odd-degree thresholds. An elementary packing argument then yields the general formula, resolving the two conjectures. The new fixed-order argument, including the shortest-odd-cycle step, and the packing normalization are checked in Lean. The final deductions are conditional on explicit published inputs.

1 Introduction

For integers $d \geq 2$ and $m \geq 0$, let $f_\Delta(d, m)$ be the maximum number of edges in a finite simple triangle-free graph G satisfying

$$\Delta(G) \leq d, \qquad \nu(G) \leq m,$$

where $\nu(G)$ denotes matching number. The main theorem gives this maximum without an implicitly defined threshold parameter.

Theorem 1.1 (General extremal formula)**.** *For $d \geq 2$, define*

$$z_d = \begin{cases} \lfloor 5d/4 \rfloor, & d \text{ even}, \\ 3(d-1)/2, & d \in \{3, 5, 7\}, \\ \lfloor 5(d+1)/4 \rfloor, & d \geq 9 \text{ odd}. \end{cases} \tag{1}$$

For every $m \geq 0$, write $m = kz_d + r$, where $0 \leq r < z_d$. Then

$$f_\Delta(d, m) = dm + k \lfloor d/2 \rfloor + \max\{0, r - d + 1\}. \tag{2}$$

The value is attained by a disjoint union of threshold components, at most one additional non-star component, and d-stars.

Ahanjideh, Ekim, and Yıldız [2] proved the formula in several parameter ranges and established the component reduction on which we build. They introduced $Z(d)$, the least matching number of a triangle-free factor-critical graph that is d-regular for even d, or has one vertex of degree $d-1$ and all others of degree d for odd d. Banak, Ekim, and Taşkın [3] extended the computational results and supplied the intermediate construction family. Their Conjectures 2.1 and 2.2 assert the intermediate component value and the unrestricted extremal formula.

We integrate the short component and endpoint proofs from [2], with attribution, into the argument below. The new step is the fixed-order upper bound: a graph exceeding the proposed component value has too little total deficiency to avoid a saturated five-cycle. The cycle excess $q = 2|V(G)| - 5d$ then governs both the extension to a C_5-coloring and the final missing-edge count. This proves the intermediate bound and identifies $Z(d) = z_d$; a single packing argument completes theorem 1.1.

Thus the previously conjectured formulas are consequences of the general theorem and its component bound. The theorem gives the extremal value and examples attaining it; it does not classify every extremal graph.

2 Reduction to factor-critical components

We omit isolated vertices. The maximum defining $f_\Delta(d, m)$ exists: the endpoints of a maximal matching form a vertex cover of size at most $2m$, so every admissible graph has at most $2md$ edges. Among graphs attaining the maximum, choose one with as many d-star components as possible. Call such a graph *normalized*. It has matching number exactly m, since otherwise adjoining a d-star would increase its edge count. Below, a non-star component means one not isomorphic to $K_{1,d}$.

We use Gallai's lemma: a connected graph whose matching number is unchanged by deletion of any vertex is factor-critical; see [2, Lemma 2.2]. The following reduction is the argument of [2, Lemma 2.4].

Lemma 2.1. *Every non-star component of a normalized extremal graph is factor-critical. It has $2j+1$ vertices, where j is its matching number, and is itself extremal for the bounds (d, j).*

Proof. If a non-star component H were not factor-critical, Gallai's lemma would give a vertex v with $\nu(H - v) \le \nu(H) - 1$. Replace H by $H - v$ and a d-star. This preserves admissibility and loses at most d edges while adding d, so the resulting graph is still extremal. It has more d-stars, a contradiction. Factor-criticality gives the asserted order. Finally, replacing H by any admissible graph with more edges would contradict extremality of the whole graph. □

The next bound is [2, Lemma 3.1]; its proof uses Mantel's theorem.

Lemma 2.2. *A connected triangle-free factor-critical graph H with matching number j has at most $j^2 + 1$ edges.*

Proof. The one-vertex case is immediate. Otherwise H is nonbipartite and has $2j+1$ vertices. Choose a shortest odd cycle of length $2s+1$, with $s \ge 2$. It is induced, and every outside vertex has at most s neighbors on it. The remaining $2(j-s)$ vertices induce at most $(j-s)^2$ edges by Mantel's theorem. Consequently

$$e(H) \le (2s+1) + 2s(j-s) + (j-s)^2 = j^2 + 2 - (s-1)^2 \le j^2 + 1.$$

□

Corollary 2.3. *Every non-star component of a normalized extremal graph has matching number at least d.*

Proof. Here $j \geq 2$: the case $j = 0$ is an omitted isolated vertex, while the only factor-critical graph on three vertices is K_3, which is not triangle-free. If $j < d$, then $j^2 + 1 < dj$, and replacing it by j copies of a d-star increases the edge count, contradicting lemma 2.2 and extremality. This is the replacement used in [2, Lemma 3.2]. □

3 Constructions, endpoints, and threshold bounds

3.1 A common construction family

We use the following consequence of Hall's theorem. A balanced bipartite graph with a vertices per side and minimum degree at least $a/2$ has a perfect matching. Indeed, a nonempty set S on one side with $|S| \leq a/2$ has at least $a/2$ neighbors. If $|S| > a/2$, every vertex on the opposite side has a neighbor in S, since otherwise its degree would be less than $a/2$.

The recipe below is the family of [3, Proposition 4.1]. We include its degree and matching verification and use the same Hall argument to establish factor-criticality throughout the stated range.

Lemma 3.1 (Intermediate construction)**.** *Let $d \geq 3$, $t \geq 0$, and $2t + 2 \leq d$. There is a triangle-free factor-critical graph $B_{d,d+t}$ with*

$$|V(B_{d,d+t})| = 2(d + t) + 1, \quad \nu(B_{d,d+t}) = d + t, \quad e(B_{d,d+t}) = d(d + t) + t + 1,$$

whose $2(d + t)$ original vertices have degree d and whose remaining vertex has degree $2t + 2$.

Proof. Start with $K_{d+t,d+t}$ on sides L, R. Partition each side into a set F of size $d - 1$ and a set H of size $t + 1$. Delete t edge-disjoint perfect matchings between F_L and F_R, delete all edges between H_L and H_R, and add a vertex v adjacent to $H_L \cup H_R$. The required disjoint matchings exist, for example by taking cyclic shifts between the two sets of size $d - 1$.

Vertices in F have degree $d + t - t = d$, those in H have degree $d - 1 + 1 = d$, and v has degree $2t + 2$. Triangle-freeness follows because the graph without v is bipartite and the neighbors of v are independent. The edge count follows by summing degrees.

Our hypotheses imply $d \geq t + 3$: this is immediate for $t = 0$, and follows from $2t + 2 \geq t + 3$ for $t \geq 1$. Deleting v leaves a balanced bipartite graph with $d + t$ vertices per side and minimum degree $d - 1 \geq (d + t)/2$. If $x \in L$ is deleted instead, match v to any vertex of H_R. The remaining bipartite graph has $d + t - 1$ vertices per side and minimum degree at least $d - 2 \geq (d + t - 1)/2$. Hall's theorem supplies a perfect matching in both cases; deletion from R is symmetric. Thus the graph is factor-critical and has matching number $d + t$. □

3.2 The left endpoint

Proposition 3.2. *For $d \geq 2$,*

$$f_\Delta(d, m) = dm \quad (0 \leq m < d), \qquad f_\Delta(d, d) = d^2 + 1.$$

Proof. If $m < d$, corollary 2.3 leaves only stars in a normalized extremal graph. If $m = d$, there is at most one non-star component, and it has matching number d. Its edge count is at most $d^2 + 1$ by lemma 2.2. These bounds are attained by m stars, and by $B_{d,d}$ from lemma 3.1, respectively. For $d = 2$, the latter construction is replaced by C_5. These are the endpoint arguments of [2, Theorems 3.3, 3.4]. □

3.3 The threshold and the upper limit on component size

We retain one construction theorem from [2, Proposition 4.3]: for every $d \geq 2$ there is a triangle-free factor-critical graph with the threshold degree profile and matching number

$$U(d) = \begin{cases} \lfloor 5d/4 \rfloor, & d \text{ even}, \\ \lfloor 5(d+1)/4 \rfloor, & d \text{ odd}. \end{cases}$$

These are the graphs denoted B_d in that paper. We use their existence, not a claim that they realize the least threshold in every degree. This is the only problem-specific construction theorem used below whose proof is not reproduced in this draft. It follows that $Z(d)$ exists. Let C_d be a graph attaining this least threshold. Its degree profile gives

$$e(C_d) = dZ(d) + \lfloor d/2 \rfloor .$$

Proposition 3.3. *For all $d \geq 2$,*

$$d \leq Z(d) \leq U(d) < 2d.$$

Moreover, $Z(d) \geq d+1$ for $d \geq 4$, $Z(d) = \lfloor 5d/4 \rfloor$ for even d, and $Z(3) = 3$, $Z(5) = 6$.

Proof. A threshold graph of matching number $j < d$ would have $dj + \lfloor d/2 \rfloor > dj$ edges, contrary to proposition 3.2. If $d \geq 4$, one of matching number d would have more than $d^2 + 1$ edges, also impossible. This proves the lower bounds; existence of the cited construction gives $Z(d) \leq U(d) < 2d$.

For even d, apply the Andrásfai–Erdős–Sós theorem in the form that every triangle-free graph on n vertices with minimum degree greater than $2n/5$ is bipartite; see [1] and [2, Corollary 2.6]. The factor-critical threshold graph is nonbipartite and d-regular, so $5d \leq 4Z(d) + 2$. Since d is even, this gives $Z(d) \geq \lfloor 5d/4 \rfloor$, as in [2, Lemma 4.7]. Finally, $B_{3,3}$ and $B_{5,6}$ from lemma 3.1 have the required almost-regular profiles, attaining the corresponding lower bounds. □

Proposition 3.4 (Component reduction)**.** *For every $d \geq 2$ and $m \geq 0$, an extremal graph can be chosen with each component either a d-star or a factor-critical extremal graph of matching number j satisfying $d \leq j \leq Z(d)$.*

Proof. Only the upper limit remains after lemma 2.1 and corollary 2.3. If a non-star component H has matching number $j > Z(d)$, then

$$e(H) \leq dj + \lfloor d/2 \rfloor .$$

Replacing it by C_d and $j - Z(d)$ stars attains that many edges and increases the number of star components, contradicting normalization. This is the argument of [2, Lemma 4.5]. □

Proposition 3.5. *For every $d \geq 2$,*

$$f_\Delta(d, Z(d)) = dZ(d) + \lfloor d/2 \rfloor .$$

Proof. The graph C_d attains the stated value. Since $Z(d) < 2d$, a normalized extremal graph has at most one non-star component. If its matching number is j, degree summation bounds its edges by $dj + \lfloor d/2 \rfloor$; the stars contribute at most $d(Z(d) - j)$. If there is no such component, the bound is immediate. This is the threshold case of [2, Theorem 4.8]. □

3.4 Deficiency notation

For the new fixed-order argument, write

$$i = d + t, \qquad n = 2i + 1 = 2d + 2t + 1, \qquad L = di + t + 1 = d(d + t) + t + 1.$$

For any graph with maximum degree at most d, put

$$\epsilon(v) = d - \deg(v), \qquad D(G) = \sum_v \epsilon(v) = dn - 2e(G).$$

Then

$$e(G) \le L \quad \Longleftrightarrow \quad D(G) \ge d - 2t - 2. \tag{3}$$

The deficiency is nonnegative and has the same parity as d, because n is odd.

4 A saturated five-cycle

The following symmetrization supplies the saturated cycle needed by the structural argument.

Lemma 4.1 (Saturated five-cycle)**.** *Let $d \ge 7$ and $1 \le t < d$. Suppose that G is a triangle-free graph on $n = 2d + 2t + 1$ vertices, with $\Delta(G) \le d$. If $e(G) > L$, then G contains an induced 5-cycle all of whose vertices have degree d.*

Proof. Since the edge count is integral, the counterexample assumption and equation (3) give

$$D(G) \le d - 2t - 4. \tag{4}$$

If G is bipartite, its smaller part has at most $d + t$ vertices. Summing degrees over this part gives $e(G) \le d(d + t) < L$, a contradiction. Thus G is not bipartite.

Let C be a shortest odd cycle, of length ℓ. It is induced. Every vertex outside C has at most two neighbors on C. Indeed, suppose that an outside vertex has at least three such neighbors. The cyclic gaps between successive neighbors have length at least two, because G is triangle-free. At least one gap is odd, and among at least three gaps such an odd gap has length at most $\ell - 4$. That gap together with the two incident edges from the outside vertex gives a shorter odd cycle.

Every vertex of G therefore sends at most two incidences into C, and hence

$$\sum_{v \in C} \deg(v) \le 2n.$$

It follows that

$$D(G) \ge \sum_{v \in C} (d - \deg(v)) \ge \ell d - 2n.$$

If $\ell \ge 7$, then

$$D(G) \ge 3d - 4t - 2 > d - 2t - 4,$$

where the strict inequality is $2(d - t + 1) > 0$. This contradicts equation (4). Triangle-freeness excludes $\ell = 3$, so G contains a 5-cycle.

Choose a 5-cycle

$$C = c_0c_1c_2c_3c_4c_0$$

maximizing $\sum_{j=0}^{4} \deg(c_j)$. Indices are read modulo five. Every outside vertex has at most two neighbors on C. Let R be the set of outside vertices with fewer than two neighbors on C, and put

$$q = \sum_{x \in R} (2 - |N(x) \cap C|), \qquad r = |R|.$$

Thus $r \le q$. Let D_C and D_{out} be the sums of the deficiencies on and outside C. Counting incidences with C gives

$$D_C = 5d - 2n + q, \qquad q + D_{\text{out}} = D(G) - 5d + 2n \le 2t - 2. \tag{5}$$

For each j, let A_j consist of c_j and every outside vertex whose two neighbors on C are c_{j-1} and c_{j+1}. These five sets partition $V(G) \setminus R$. Write

$$a_j = |A_j|, \qquad b_j = |N(c_j) \cap R|.$$

The definition of the bags gives

$$\deg(c_j) = a_{j-1} + a_{j+1} + b_j.$$

Adding this identity for $j+2$ and $j+3$ yields

$$\begin{aligned} a_j &= n - r - \deg(c_{j+2}) - \deg(c_{j+3}) + b_{j+2} + b_{j+3} \\ &\ge 2t + 1 - r. \end{aligned} \tag{6}$$

Consequently,

$$|A_j \setminus \{c_j\}| \ge 2t - r > D_{\text{out}},$$

where the strict inequality follows from equation (5) and $q \ge r$. Every vertex of degree less than d contributes at least one to D_{out}. Therefore $A_j \setminus \{c_j\}$ contains a vertex x_j of degree d. Replacing c_j by x_j produces another 5-cycle. The maximal choice of C now implies $\deg(c_j) = d$. This holds for every j. □

The bags A_j are independent. Moreover, edges between distinct bags can occur only between consecutive bags: vertices in any forbidden pair share a neighbor on C, so an edge between them would create a triangle.

5 Stability around a saturated five-cycle

The natural parameter is the *cycle excess*

$$q = 2n - 5d.$$

For a saturated five-cycle, this is exactly the number of missing incidences into the cycle:

$$q = \sum_{x \in R} (2 - |N(x) \cap C|).$$

In particular, $q \ge 0$. The original bags already form a subgraph of a blow-up of C_5; the question is whether the exceptional vertices can be assigned to the same five positions.

We prove a structural statement for arbitrary order n, independent of the matching-number problem. Put

$$T = 8d - 3n.$$

This quantity is the deficiency forced by an eight-vertex subgraph with independence number three: each graph vertex sends at most three incidences into that subgraph.

Theorem 5.1 (Saturated-cycle stability)**.** *Let G be a triangle-free graph on n vertices with $\Delta(G) \le d$, containing a five-cycle $C = c_0c_1c_2c_3c_4c_0$ whose vertices all have degree d. Set $D = dn - 2e(G)$, $q = 2n - 5d$, and $T = 8d - 3n$. If either*

$$D < T, \qquad \text{or} \qquad D \le T, \quad T > 0, \quad q \ge 2,$$

then there is a homomorphism $G \to C_5$ taking c_j to j.

The saturated-cycle hypothesis is essential to the scope of this statement: we do not claim here that every graph near the extremal edge count has such a cycle. The theorem itself has examples, such as balanced complete blow-ups of C_5, for which $D = q = 0$ and $T > 0$.

5.1 The two local obstructions

Two ways of failing to extend the bag assignment lead to the same graph H. It has distinct vertices $a, b, u, v, p_1, p_2, x_1, x_2$, the four edges between $\{a, b\}$ and $\{u, v\}$, and the two paths

$$a - p_1 - p_2 - b, \qquad u - x_1 - x_2 - v.$$

Its independence number is three. An independent set cannot meet both $\{a, b\}$ and $\{u, v\}$. If it meets their union once, the edges p_1p_2 and x_1x_2 allow at most two further vertices; if it avoids the union, they allow at most two. If it contains both a, b, it excludes p_1, p_2 and contains at most one of x_1, x_2; the case of u, v is symmetric.

A triangle-free supergraph of H on these eight vertices may add edges only between $\{p_1, p_2\}$ and $\{x_1, x_2\}$, and the added edges form a matching. Adding two gives the eight-vertex Möbius ladder M, the graph on $\mathbb{Z}/8\mathbb{Z}$ with differences 1, 4, or 7 adjacent. Thus M arises as the unique two-edge augmentation, up to isomorphism, of the obstruction to extending the coloring.

The graph M is triangle-free and 3-regular. Its independence number is three, and its independent triples are precisely the neighborhoods

$$\{j - 1, j + 1, j + 4\}, \qquad j \in \mathbb{Z}/8\mathbb{Z}.$$

Indeed, an independent four-set in the 8-cycle must be alternating, and both alternating sets contain an opposite chord. For an independent triple, the cyclic gaps are at least two and sum to eight. The gap pattern $(2, 2, 4)$ contains an opposite chord; $(2, 3, 3)$ gives the displayed neighborhoods. Distinct nonadjacent vertices of M have a common neighbor.

Lemma 5.2 (Obstruction cost and equality)**.** *Every copy of M in G forces $D \ge T$. If $T > 0$ and $q \ge 2$, equality $D = T$ is impossible.*

Proof. Let S be the eight vertices of a copy. Every $N(x) \cap S$ is independent in M, so has size at most three. Therefore

$$D_S := \sum_{v \in S} (d - \deg_G(v)) \ge 8d - 3n = T.$$

If $D = T$, equality holds throughout: all deficiency lies on S, every vertex outside S has degree d, and every vertex has exactly three neighbors in S. In particular, $G[S] = M$.

Partition $V(G)$ into bags B_j according to the neighborhood $\{j-1, j+1, j+4\}$ in S, with $j \in B_j$. Put $a_j = |B_j|$ and $\delta_j = d - \deg_G(j)$. The common-neighbor property makes every bag independent and permits edges between bags only along edges of M. Also

$$\deg_G(j) = a_{j-1} + a_{j+1} + a_{j+4}.$$

Adding this identity at $j-1, j, j+1$ counts every bag once and B_j twice, so

$$a_j = 3d - n - (\delta_{j-1} + \delta_j + \delta_{j+1}) \geq 3d - n - D_S = 2n - 5d = q.$$

If $q \geq 2$, each bag contains a vertex outside S. That vertex is saturated and has no more possible neighbors than j, forcing $\deg_G(j) = d$. Thus all of S is saturated, contradicting $D_S = T > 0$. □

Lemma 5.3. *If G contains no M, every copy of H forces $D \geq T + 2$.*

Proof. On the vertex set S of a copy of H, at most one edge can be added: two added edges would give M. Hence $e(G[S]) \leq 11$. Every outside vertex has at most three neighbors in S, giving

$$\sum_{v \in S} \deg_G(v) \leq 22 + 3(n-8) = 3n - 2.$$

Consequently $D \geq D_S \geq 8d - 3n + 2 = T + 2$. □

5.2 Extending the cycle coloring

Lemma 5.4 (Extension criterion)**.** *If G contains a saturated five-cycle, contains no H, and $D \leq T$, then its natural bag assignment extends to a homomorphism $G \to C_5$.*

Proof. Use the bags A_j and exceptional set R of section 4, and write $r = |R|$ and $b_j = |N(c_j) \cap R|$. Saturation gives

$$r \leq q, \qquad \sum_j b_j = 2r - q.$$

The exact bag identity therefore yields the uniform bound

$$a_j = n - r - 2d + b_{j+2} + b_{j+3} \leq n - 2d. \tag{7}$$

Indeed, $b_{j+2} + b_{j+3} \leq 2r - q \leq r$.

No $x \in R$ can have neighbors in consecutive bags. Otherwise, after relabeling, choose $u \in A_0$ and $v \in A_1$ adjacent to x. There is a copy of H with

$$(a, b, u_H, v_H, p_1, p_2, x_1, x_2) = (c_0, c_2, v, c_1, c_4, c_3, x, u).$$

The two paths are $c_0 - c_4 - c_3 - c_2$ and $v - x - u - c_1$, and the four cross-edges come from the bag definitions. The vertices are distinct: $u = c_0$ or $v = c_1$ would create a triangle with x.

For every $x \in R$, the deficiency bound gives

$$|N(x) \setminus R| \geq d - D - r + 1 \geq d - T - q + 1 = n - 2d + 1. \tag{8}$$

This exceeds the size of any one bag. Its occupied positions form an independent set in C_5, so x meets exactly two nonconsecutive bags. Assign x to their unique common neighboring position.

Only edges inside R remain to check. If an edge xy has equal or nonadjacent assigned positions, the two occupied bag-pairs share a bag A_j. Choose $u \in N(x) \cap A_j$ and $v \in N(y) \cap A_j$. Triangle-freeness gives $u \neq v$. The vertices x, y, u, v and the four cycle vertices other than c_j contain H, with

$$(a, b, u_H, v_H, p_1, p_2, x_1, x_2) = (c_{j-1}, c_{j+1}, u, v, c_{j-2}, c_{j+2}, x, y).$$

They are distinct even if u or v equals c_j, since c_j was omitted. This contradiction proves that every edge maps to an edge of C_5. □

Proof of theorem 5.1. Under either hypothesis, lemma 5.2 excludes M. Since $D \leq T$, lemma 5.3 then excludes H, and lemma 5.4 extends the coloring. □

6 Odd excess costs deficiency

Once the coloring is available, the excess has a second interpretation. Let $B_0, \dots, B_4$ be its color classes, each containing the corresponding saturated cycle vertex, and set

$$s_j = |B_j|, \qquad p_j = s_{j-1} + s_{j+1} - d, \qquad h = n - 2d.$$

Saturation ensures $p_j \geq 0$. Direct summation gives

$$\sum_j p_j = q, \qquad s_j = h - p_{j+2} - p_{j+3}. \tag{9}$$

Thus the same excess q counts both the original missing cycle incidences and the surplus neighboring capacity after coloring.

Let F be the complete blow-up on these classes, put $\mu = e(F) - e(G)$, and write $w_j = s_j p_j$ and $W = \sum_j w_j$. Every vertex in B_j has $d + p_j$ possible neighbors, so

$$D = 2\mu - W. \tag{10}$$

For any independent set I of cycle positions, a missing edge has at most one endpoint in its classes. Since at least w_j incidences are missing from B_j,

$$D \geq 2\sum_{j \in I} w_j - W. \tag{11}$$

The following argument replaces the capacity table. It applies more generally whenever the positions with positive excess have bipartite support.

Lemma 6.1 (Odd excess imbalance)**.** *Suppose p_j are nonnegative integers with odd sum $q = 2k-1 > 0$, and the subgraph of C_5 on $\{j : p_j > 0\}$ is bipartite. If $h \geq q$ and $w_j = p_j(h - p_{j+2} - p_{j+3})$, then some independent set I satisfies*

$$2\sum_{j \in I} w_j - \sum_j w_j \geq h - k(k-1).$$

In particular, for $q = 1$ or $q = 3$, equation (11) gives $D \geq h$ or $D \geq h - 2$, respectively.

Proof. Bipartition the positive support into independent sets I, J, choosing I to have larger total p-weight. Put $P = \sum_{j \in I} p_j$; integrality and oddness give $k \leq P \leq q$.

Expand the difference of the two w-weights. Products $p_i p_j$ from nonadjacent positions on opposite sides cancel. Those within J have positive sign and may be discarded. Those within I cost at most

$$2\sum_{\{i,j\} \subseteq I} p_i p_j = P^2 - \sum_{i \in I} p_i^2 \leq P(P-1),$$

because the p_i are integers. Therefore

$$2\sum_{j \in I} w_j - W \geq h(2P - q) - P(P-1) \geq h - k(k-1).$$

The last inequality follows from

$$h(2P - q) - P(P-1) - (h - k(k-1)) = (P-k)(2h - P - k + 1) \geq 0,$$

using $P \geq k$ and $P, k \leq q \leq h$. For $q \leq 3$, the positive support has at most three positions, so it is automatically bipartite. □

7 The fixed-order bound

We now apply the structural and capacity statements with $n = 2d+2t+1$. The relevant parameters are

$$q = 4t + 2 - d, \qquad h = 2t + 1, \qquad T = 2d - 6t - 3. \tag{12}$$

Proposition 7.1. *Let $d \geq 7$ and $1 \leq t < d$. If $q = 4t + 2 - d$ is negative, one, or three, then every triangle-free graph on $2d + 2t + 1$ vertices with maximum degree at most d has at most $L = d(d + t) + t + 1$ edges.*

Proof. A counterexample has a saturated five-cycle by lemma 4.1. Hence $q \geq 0$, disposing of negative excess. Moreover,

$$D \leq d - 2t - 4 = T + q - 3. \tag{13}$$

If $q = 1$, then $D \leq T - 2 < T$. If $q = 3$, then $d = 4t - 1$ and $D \leq T = 2t - 5$. Nonnegativity of D forces $t \geq 3$, so $T > 0$. In either case theorem 5.1 gives a homomorphism to C_5 preserving the saturated cycle. Since $h = 2t + 1 \geq q$, the odd excess imbalance gives, in both cases,

$$D \geq h - (q - 1) = 2t + 2 - q.$$

But equation (13) also reads $D \leq 2t - 2 - q$, a contradiction. □

8 Determining the threshold

We identify the least threshold before using it in the general formula. The following argument applies the fixed-order proposition directly, without assuming the desired value of $Z(d)$.

Proposition 8.1. *For odd $d \geq 9$,*

$$Z(d) = \left\lfloor \frac{5(d+1)}{4} \right\rfloor.$$

Proof. The upper bound is proposition 3.3. Suppose a threshold graph occurs at matching number $j < U(d)$. The same proposition gives $j \geq d+1$. Write $j = d+t$. Since $j < U(d)$, we have $4t \leq d+1$ and $t < d$. The odd integer $q = 4t + 2 - d$ is at most three, so it is negative, one, or three. By proposition 7.1, the graph has deficiency at least

$$d - 2t - 2 \geq (d - 5)/2 \geq 2.$$

Its threshold degree profile has deficiency one, a contradiction. □

Lemma 8.2. *We have $Z(7) = 9$.*

Proof. The construction $B_{7,9}$ from lemma 3.1, with $t = 2$, is triangle-free and factor-critical, with eighteen vertices of degree seven and one of degree six. It therefore gives $Z(7) \leq 9$ directly.

For the reverse bound, proposition 3.3 gives $Z(7) \geq 8$. A threshold graph of matching number eight would have seventeen vertices and $(7 \cdot 17 - 1)/2 = 59$ edges. But proposition 7.1, with $d = 7, t = 1$, allows at most $7 \cdot 8 + 2 = 58$. Thus $Z(7) = 9$. □

Proposition 8.3 (Explicit threshold)**.** *For every $d \geq 2$, $Z(d) = z_d$, where z_d is defined in equation (1).*

Proof. Combine the even-degree and small odd values of proposition 3.3 with proposition 8.1 and lemma 8.2. □

9 The intermediate component bound

Theorem 9.1. *Let $d \geq 7$ and $d < i < Z(d)$. Every triangle-free graph on $2i+1$ vertices with maximum degree at most d has at most $di + i - d + 1$ edges.*

Proof. Write $i = d + t$; proposition 3.3 gives $1 \leq t < d$. If d is even, $i < \lfloor 5d/4 \rfloor$ implies $4i \leq 5d - 4$, so $q = 4i + 2 - 5d \leq -2$. If d is odd, $i < \lfloor 5(d+1)/4 \rfloor$ implies $4i \leq 5d + 1$, so $q \leq 3$ and q is odd. In either case proposition 7.1 applies. □

Proposition 9.2 (Intermediate value)**.** *For $7 \leq d < i < Z(d)$,*

$$f_\Delta(d, i) = di + i - d + 1.$$

Proof. Write $i = d + t$ and use the normalized graph of proposition 3.4. Since $i < 2d$, there is at most one non-star component. If there is none, there are at most di edges. Otherwise let its matching number be $j = d + s \leq i$. For $s = 0$, proposition 3.2 gives at most $dj + 1$ edges; for $s > 0$, theorem 9.1 gives at most $dj + s + 1$. Together with the stars this is at most

$$di + s + 1 \leq di + t + 1.$$

For attainment, the bounds $i < U(d)$ imply $2t + 2 \leq d$ when $d \geq 7$. Explicitly, $4t \leq d - 4$ for even d, and $4t \leq d + 1$ for odd d. Thus lemma 3.1 supplies $B_{d,d+t}$ with the stated value. □

Remark 9.3. The fixed-order theorem requires no factor-criticality assumption. The component reduction uses this stronger conclusion only on odd-order factor-critical components.

10 The general formula as a packing theorem

For $d \geq 4$, the component bounds above reduce all matching bounds to one packing problem. Set

$$z = Z(d), \qquad b = \lfloor d/2 \rfloor .$$

The explicit threshold values in proposition 8.3 imply

$$d < z < 2d, \qquad z \leq d + b - 1. \tag{14}$$

Relative to the baseline dm, a component of matching number j has utility

$$u(j) = \begin{cases} j - d + 1, & d \leq j < z, \\ b, & j = z. \end{cases}$$

At $j = d$, the first line follows from proposition 3.2; for $d < j < z$, it follows from proposition 9.2. This interval is empty for $d = 4, 5, 6$, since then $z = d+1$. The second line follows from proposition 3.5. Stars have zero utility and fill unused matching capacity. This reduction is the component-packing viewpoint of [2, Lemma 4.9]; the normalization below proves the needed optimization directly.

Lemma 10.1 (Packing normalization)**.** *Any feasible list of component weights in $[d, z]$ can be replaced, without increasing its total weight or decreasing its utility, by a list containing as many weight-z components as the capacity permits and at most one component of weight below z.*

Proof. Take two weights $a, c < z$, and put $s = a + c$. Since $s \geq 2d > z$, replace them by one weight-z component. If $s - z \geq d$, add a component of weight $s - z$; otherwise fill the remaining capacity with stars. The total weight does not increase.

If $s - z \geq d$, the utility gain is

$$b + (s - z - d + 1) - (s - 2d + 2) = b + d - z - 1 \geq 0$$

by equation (14). If $s - z < d$, then $s \leq z + d - 1$ and the old utility is

$$s - 2d + 2 \leq z - d + 1 \leq b.$$

Thus the replacement never decreases utility and strictly decreases the number of sub-threshold components. Repetition leaves at most one.

Now write the total capacity as $m = kz + r$, with $0 \leq r < z$, and suppose the normalized list contains $k' < k$ weight-z components. Discard its one smaller component if present and add one weight-z component. The new weight is $(k' + 1)z \leq kz \leq m$, and the utility does not decrease because $u(j) \leq b$ for every $j < z$. Repeating gives exactly k threshold components. The remaining capacity is r, in which at most one residual component can be placed. □

Proof of theorem 1.1. First proposition 8.3 identifies $Z(d) = z_d$.

For $d = 2, 3$, this threshold equals d. By proposition 3.4, every non-star component of a normalized extremal graph has matching number d, and contributes at most $d^2 + \lfloor d/2 \rfloor$ edges. Hence at most $\lfloor m/d \rfloor$ such components can occur, giving

$$f_\Delta(d, m) \leq dm + \lfloor m/d \rfloor \lfloor d/2 \rfloor .$$

This is attained by that many copies of C_d and the remaining stars. It agrees with equation (2), since $r < z_d = d$.

Now let $d \geq 4$ and $m = kz_d + r$. The component reduction and lemma 10.1 give k threshold components and at most one residual non-star component in an optimal packing. If $r < d$, the residual utility is zero. If $r \geq d$, it is at most $r - d + 1$. Therefore

$$f_\Delta(d, m) \leq dm + k \lfloor d/2 \rfloor + \max\{0, r - d + 1\}.$$

For attainment, take k copies of C_d. If $r < d$, add r stars. If $r = d$, add the left-endpoint graph from proposition 3.2. If $r > d$, use $B_{d,r}$ from lemma 3.1. This last case can occur only for $d \geq 7$, and the parameter check in proposition 9.2 applies. These graphs have matching number m and attain the upper bound, completing the proof for every $m \geq 0$. □

11 Consequences

Corollary 11.1 (The two conjectures)**.** *Conjectures 2.1 and 2.2 of [3] hold: the intermediate value is $f_\Delta(d, i) = di + i - d + 1$ for $7 \leq d < i < Z(d)$, and the general formula in terms of $Z(d)$ holds for every $d \geq 2$ and $m \geq 0$.*

Proof. The first assertion is proposition 9.2. The second is theorem 1.1 with $z_d = Z(d)$ from proposition 8.3. □

Corollary 11.2. *Let $d \geq 2$ and $d < m < Z(d)$. Every extremal triangle-free graph G with $\Delta(G) \leq d$, $\nu(G) = m$, and $|V(G)| = 2m + 1$ is factor-critical.*

Proof. If G is not factor-critical, there is a vertex v such that $G - v$ has no perfect matching. Since $G - v$ has $2m$ vertices, its matching number is at most $m - 1$. Therefore

$$e(G) \leq f_\Delta(d, m-1) + d = f_\Delta(d, m) - 1,$$

where the equality follows from proposition 9.2, including the known left endpoint when $m = d + 1$. This contradicts extremality. □

12 Verification status and points for review

The argument above is a proof draft and has not yet been independently refereed. Lean 4.33.0 with Mathlib checks the following parts:

- the graph argument from shortest odd cycles through the saturated 5-cycle, the M and H exclusions, the C_5-coloring, and every fixed-order degree case;
- the required capacity inequalities, the general stability theorem, and the terminating packing normalization;
- boundedness and attainment for $f_\Delta(d, m)$ and $Z(d)$, the full Proposition 4.1 family $B_{d,d+t}$ (order, triangle-freeness, degrees, matching number, and edge count), and the factor-critical graph $B_{7,9}$ used for the degree-seven threshold.

The integrated mathematical proof above includes the short published component and endpoint arguments. The Lean development still represents those results, the general threshold constructions, the disjoint-union lower construction for the full formula, and the previously solved small cases through the explicit interface `AuditedLiteratureInputs`. Their inclusion as prose proofs here does not discharge those formal hypotheses. From that interface, Lean proves the fixed-order upper bound and the packing deduction, and derives both conjectures in terms of $Z(d)$. Proposition 4.1 is no longer assumed by that interface: Lean constructs its graph and derives the residual lower bounds directly; the remaining disjoint-union field takes those bounds as premises. The complete identification $Z(d) = z_d$ in proposition 8.3 is proved in the manuscript. Lean additionally proves the degree-seven case and proposition 8.1 for every odd $d \geq 9$, the latter from the published bounds $d + 1 \leq Z(d) \leq \lfloor 5(d+1)/4 \rfloor$. The general Hall proof of factor-criticality in lemma 3.1 remains a written proof.

The complete Lean build audits the named theorems listed in `Audit.lean`. It contains no admitted proof, project-specific axiom, or native-evaluation shortcut. The reported dependencies are only Lean's standard foundational principles ("propext", classical choice, and quotient soundness). Lean formalizes the weighted bipartition proof of lemma 6.1 and uses it for the excess-one and excess-three capacity conclusions. The earlier finite pattern proofs remain as independent regression checks. For $B_{7,9}$, Lean checks explicit matching partners after each of the nineteen vertex deletions, in place of the Hall argument given above. The upper bound $Z(7) \leq 9$ has no external hypotheses; its lower bound uses only the general published lower bound $8 \leq Z(7)$. The formal files and the exact map from assumptions to published theorem numbers will accompany a later version of this draft.

The main points for mathematical review are the transition from normalized components to the fixed-order problem, lemma 4.1, the two obstruction embeddings in section 5, the odd excess imbalance, the threshold identification in section 8, and the packing replacement in lemma 10.1. The attribution of the integrated published arguments is recorded where each is used.

Remark 12.1 (A discrepancy in the 2023 preprint). The introductory discussion of [3] gives $Z(9) = 13$ once, whereas its Section 4 and computational table give $Z(9) = 12$. The latter is consistent with the proven upper bound $Z(9) \leq \lfloor 5(9+1)/4 \rfloor = 12$ from [2]. No argument here uses the value 13. There is a second table inconsistency: Table 5 reports 148 edges for $(d, m) = (8, 18)$, while Table 6 reports 149. The latter agrees with the conjectured formula and with the displayed decomposition into one $Z(8)$-component and one component of matching number eight.

Verification files

The Lean development, the finite verification script, and the map from Lean inputs to published results are available at https://github.com/erdembanak/extremal-IP/tree/master/proof.

References

[1] B. Andrásfai, P. Erdős, and V. T. Sós, On the connection between chromatic number, maximal clique and minimal degree of a graph, *Discrete Mathematics* 8 (1974), 205–218.

[2] M. Ahanjideh, T. Ekim, and M. A. Yıldız, Maximum size of a triangle-free graph with bounded maximum degree and matching number, *arXiv preprint arXiv:2207.02271*, 2022. https://arxiv.org/abs/2207.02271.

[3] A. E. Banak, T. Ekim, and Z. C. Taşkın, Constructing extremal triangle-free graphs using integer programming, *arXiv preprint arXiv:2304.01729*, 2023. https://arxiv.org/abs/2304.01729.